DON-TSA DELCHERE

Computational Physicist | AI & Machine Learning | Energy Materials

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PROFILE

Researcher specializing in AI-driven computational modeling and nanoscale simulations, with a focus on accelerating the discovery of energy materials for sustainable development. Currently completing a PhD, developing machine learning frameworks to solve complex physical systems challenges. Committed to harnessing AI to advance Africa's ecological transition and drive impactful, data-informed solutions at the intersection of science, technology, and society.

EDUCATION

PhD in Physics — 2022 – Present

University of Lomé, Togo

Doctoral research on accelerating energy material identification through nanoscale simulations and machine learning. Developing predictive AI models combining first-principles calculations with data-driven approaches.

MSc. Mathematical Sciences — 2021 – 2022

African Institute for Mathematical Sciences (AIMS), Cameroon

Advanced training in mathematics, quantum computing, and data science. Strong foundations in ML, AI, and quantitative research methodologies.

MSc. Condensed Matter Physics — 2019 – 2021

University of Dschang & University of Lorraine, France

Theoretical and computational physics: quantum mechanics, field theory, nanoscale matter. Deep expertise in semiconductor physics and nanomagnetism.

BSc. Physics — 2016 – 2019

University of Dschang, Cameroon

Classical and modern physics: mechanics, electromagnetism, thermodynamics, and quantum physics.

RESEARCH EXPERIENCE

Visiting Research Fellow — Jun – Jul 2025

Ștefan cel Mare University of Suceava — Eugen Ionescu Scholarship — Romania

- Explored new computational approaches to energy materials research within a European academic setting.
- Expanded academic network across European research institutions.

Visiting Research Fellow — May – Jul 2024

Imperial College London — Global Development Hub Fellows Fund — London, UK

- Leveraged HPC infrastructure for advanced nanoscale simulations, significantly accelerating research capacity.
- Acquired new simulation methodologies and applied them directly to materials research projects.
- Engaged with a world-class international research community, broadening scientific perspectives.

Mentor — Big Data & Machine Learning — Jun – Nov 2023

iAnalyse Mentorship Program, Cohort 1 — Remote

- Delivered courses on dimensionality reduction, data cleaning, and machine learning models.
- Prepared learners to tackle real-world data challenges; deepened technical and pedagogical skills.

PUBLICATION

DON-TSA, D. et al. (2024). *"Predictive Models for Inorganic Materials Thermoelectric Properties with Machine Learning."* *Machine Learning: Science and Technology*, IOP Publishing. DOI: 10.1088/2632-2153/ad6831

TECHNICAL SKILLS

Programming	Python, Bash/Linux, LaTeX
Simulation & Modelling	VASP, Quantum ESPRESSO, DFT, First-Principles Calculations, Nanoscale Simulation, Physics-Informed Neural Networks (PINNs)
ML & Data Science	Scikit-learn, TensorFlow, PyTorch, Keras, Large Language Model, Prompt Engineering, RAG, AI Agents, Agent Workflows, XGBoost, Pandas, NumPy, Matplotlib, Supervised & Unsupervised Learning, Regression, DBSCAN Clustering, Data Preprocessing,LangChain
Tools & Platforms	Jupyter Notebook, Google Colab, Git/GitHub, HPC Clusters (SLURM), Data Analytics

LANGUAGES

Language	Proficiency	CEFR Level
French	Native / Bilingual Proficiency	**C2 — Mastery**
English	Full Professional Proficiency	**C1 — Advanced**

CONFERENCES & PROFESSIONAL DEVELOPMENT

Deep Learning Indaba — *Kigali, Rwanda* **Aug 2026**
Deep learning methods for scientific research; African ML community engagement.

AI Tech Industry Day — *Lomé, Togo* **Feb 2026**
Cross-sector event bridging laboratory innovation and market application.

Google Data Analytics Professional Certificate — *Google / Coursera* **Jul 2025**

Deep Learning Indaba — *Kigali, Rwanda* **Aug 2025**
Deep learning methods for scientific research; African ML community engagement.

ICTP PWF-TOGO 2024 — *Lomé, Togo* **Aug 2024**
Tutor for materials modeling workshops.

F60 TYC — 7th Energy Materials Workshop — *London, UK* **Jul 2024**
Modelling Energy Materials.

TYC Postgraduate Student Day 2024 — *Imperial College London, UK* **May 2024**
Poster: predictive models for thermoelectric properties using ML.

7th ASESMA — *Kigali, Rwanda* **Jun 2023**
Material composition and structure as features for ML models.

HPC for Sustainable Development (SMR 3828) — *Stellenbosch, South Africa* **Apr 2023**

SOFT SKILLS

Scientific Writing & Publication · Science Communication & Outreach · Mentoring & Teaching · Multidisciplinary Team Collaboration · Poster & Conference Presentation · Storytelling for Technical Audiences

REFERENCES

Available upon request.